



SWMS06

Safe Work Method Statement

Work in or near a confined space

Document number	EC-SWMS06	Revision	1.0
Prepared by	Ecology Consulting — WHS / Field team	Document release date	
Approved by	Tony Webster — Managing Director / WHS Officer	Next review	

Note: Work must be performed in accordance with this Safe Work Method Statement (SWMS). This SWMS must be kept and available for inspection until the work is completed. If revised, every version must be kept. If a notifiable incident occurs in relation to the work in this SWMS, the SWMS must be kept for at least two years from the date of the notifiable incident. This SWMS is accessed and acknowledged through the Ecology Consulting digital WHS platform; field forms are submitted from site and reviewed centrally by the administration team.

Scope & limitations: Culverts, stormwater drains, bridge voids, underpass cavities, pits, pipes and enclosed waterways are treated as confined spaces until proven otherwise. **Ecology Consulting personnel will not enter a confined space unless a confined space entry permit system is implemented and authorised by the Principal Contractor, and the entrant holds current confined space entry competency.** Where entry is not permitted, works are limited to inspection from the access point using cameras, torches and borescopes. Ecology Consulting personnel will not undertake confined space rescue under any circumstances.

Project & PCBU details

Person conducting a business or undertaking (PCBU)	Ecology Consulting Pty Ltd · ABN 68 647 633 163	Contact number	0472 569 988 (office / Check-in)
EC offices	34 Atchison St, Wollongong NSW 2500 · 42 Goulburn St, Crookwell NSW 2583 · 473 Townsend St, Albury NSW 2640		
Client / Principal Contractor			
Work activity	Ecological survey and fauna assessment in or near confined spaces including culverts, stormwater drains, bridge voids, underpass cavities, pits, pipes and enclosed waterways.		
Workplace location			

Field work duration	
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Legislation, codes of practice & standards

Acts & Regulations	Work Health and Safety Act 2011 (NSW) and the Work Health and Safety Regulation (NSW); Work Health and Safety Act 2011 (Cth) and WHS Regulations 2011 (Cth) where work is on Commonwealth land or for a Commonwealth entity (Comcare jurisdiction); Work Health and Safety Act 2011 (ACT) and WHS Regulation 2011 (ACT) for works in the ACT.
Codes of Practice	Safe Work Australia model Codes as adopted in NSW and the ACT: <i>How to manage work health and safety risks; Work health and safety consultation, cooperation and coordination; Construction work; First aid in the workplace; Managing the work environment and facilities; Preparation of safe work method statements for construction work. Confined spaces; Managing risks of hazardous chemicals in the workplace.</i>
Standards & guidance	AS 2865 Confined spaces; AS/NZS 1715 and AS/NZS 1716 Respiratory protective equipment; AS/NZS 1891 Industrial fall-arrest systems and devices; AS/NZS 1801 Occupational protective helmets; AS 1319 Safety signs.
EC WHS documentation	EC Work Health and Safety Management System and WHS Handbook; EC Risk Management Processes (EC-RMP-FRM-001); EC Hazard Register and Trigger Action Response List (TARL); Pre-Mobilisation Checklist; Pre-Travel Vehicle Safety Checklist; Journey Management Plan; Daily Toolbox Talk; site-specific Emergency Response Plan (ERP); JSA — General Field Surveys; JSA — Remote and Isolated Work; SOP — Journey Management; SOP — Terrestrial Fauna; SOP — Fieldwork Biosecurity.

Mandatory personal protective equipment & field equipment

Confirmed present and serviceable on the Pre-Mobilisation Checklist and verified by the Field Lead at the Daily Toolbox Talk. Faulty items are tagged out and removed from service.

Hard Hat	P2 Respirator	Safety Glasses	Gloves	Hearing Protection	Hi-Vis Clothing	Disposable Coveralls	Long Sleeves	Capped Boots
☐	☐	☐	☐	☐	☐	☐	☐	☐

Fire Extinguisher	Remote First Aid Kit	Snake Bite Kit	In Reach / Sat Phone	Two-Way Radio	Torch / Headlamp	GPS	Drinking Water
☐	☐	☐	☐	☐	☐	☐	☐

Calibrated Gas Detector	Mechanical Ventilation	Harness & Lifeline	Tripod / Retrieval System	Torch (two per person)	Entry Permit & Log	Standby Person	Non-Entry Rescue Kit
☐	☐	☐	☐	☐	☐	☐	☐

Hierarchy of controls

Eliminate	Avoid entry wherever possible — inspect from the access point using remote cameras, borescopes, torches and extendable equipment, or rely on existing data.
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Substitute	Use safer methods or equipment — non-entry fauna detection, remote lighting and monitoring, and survey from outside the space.
Isolate	Separate people from hazards — isolate water inflows and discharge sources, mechanical and electrical systems, adjacent plant and equipment, and vehicle and traffic hazards above or beside the space.
Engineer	Ventilation (natural or mechanical), physical barriers, improved access points, fixed lighting and retrieval systems provided by the Principal Contractor.
Administrative	SWMS, confined space entry permits, atmospheric testing, standby person, minimum two-person teams, entry and exit tracking, communication protocol, rescue plan, and enforced stop-work triggers.
PPE	Last line of defence — respirators, gloves, helmet, coveralls, eye protection and harness where a retrieval system is in use.

Hazards & controls

Complete for every entry. A valid permit and a current atmospheric test are required before any person enters.

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Planning, preparation and travelling to site		
Inadequate planning leading to uncontrolled risk; unfit or untrained personnel	Attend the pre-field meeting with the Project Lead and Project Manager to review scope, methodology and expected hazards; confirm site access, keys, codes, inductions and property constraints; and confirm all staff are trained, competent and fit for work. Medical concerns are disclosed privately to the Field Lead. Review this SWMS, the relevant JSA and SOP, the Field Plan and the site-specific ERP. Verify training currency — workers are not mobilised with expired mandatory training.	☐ Pre-field meeting ☐ SWMS / JSA / SOP reviewed ☐ ERP reviewed ☐ Training verified ☐ Medical disclosures
Vehicle failure, missing safety equipment or fatigue during travel	Complete the Pre-Travel Vehicle Safety Checklist — first aid kit, extinguisher, spare tyre, torch, water and PPE present; vehicle free from leaks, faults or damage; electrical issues visually inspected only and tagged out if unsafe. Complete the Journey Management Plan. Minimum 10-hour rest between shifts before driving; personal fatigue self-assessment before departure; rest breaks every two hours; carry water, snacks and electrolytes; second driver for long or remote trips where practicable. Zero blood alcohol concentration applies to all work driving.	☐ Vehicle checklist done ☐ JMP completed ☐ 10-hr rest confirmed ☐ Fatigue self-assessment
Travelling into fire, flood, storm, smoke or hazardous road conditions	Complete the Pre-Travel Environmental and Hazard Check using BOM weather warnings, Hazards Near Me, RFS fire danger ratings, Air Quality Index, lightning trackers, Live Traffic NSW and telecommunications coverage maps. Do not travel where there are emergency warning fire events, hazardous air quality, active storm warnings or flooded roads. Never drive through floodwater under any circumstance. If conditions change, stop, reassess and notify the Communication Officer.	☐ BOM / Hazards Near Me ☐ RFS rating checked ☐ AQI checked ☐ Live Traffic NSW ☐ Coverage checked
Hazards not identified, assessed or communicated before work commences	All team members participate in a pre-commencement Daily Toolbox Talk led on site by the Field Lead and documented on the EC WHS form — site-specific hazards identified; collaborative risk assessment using the EC risk matrix (rating one to six) with agreed controls and a revised residual rating; PPE and equipment readiness confirmed; ERP reviewed and Emergency Meeting Point identified; communication and check-in requirements confirmed; health or access needs disclosed privately; and worker sign-on confirming understanding and fitness for work. Hazards already in the EC Hazard Register are managed under the TARL. No worker commences without sign-on.	☐ Toolbox talk held ☐ Risk assessment rated ☐ Emergency Meeting Point ☐ All workers signed on
Loss of contact with the field team; delayed emergency response	Mandatory check-ins with the Communication Officer (0472 569 988) before departing base, on arrival at site or accommodation, before leaving site, and on return to base. Every check-in must be acknowledged. A missed or unacknowledged check-in triggers escalation through the Communication Officer, Project Lead or Project Manager, and the WHS Officer, and is grounds for activating the ERP. Carry redundant communications — mobile phone, UHF radio and In Reach or satellite communicator, with a PLB where remoteness warrants. Roles are assigned at the toolbox talk: Field Lead, Communication Officer and workers.	☐ Check-in schedule agreed ☐ Comms tested ☐ Roles assigned ☐ Escalation briefed
Determination, permits and competency		

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Unauthorised or uncontrolled entry into a confined space	Prior to any works the Field Lead must determine whether the work area meets the definition of a confined space — an enclosed or partially enclosed space that is not designed for continuous human occupancy, has limited entry or exit, and may contain hazardous atmospheres, contaminants or engulfment risks. Examples relevant to ecological survey include culverts and stormwater drains, bridge voids and underpass cavities, and pits, pipes and enclosed waterways. Where any uncertainty exists, the space must be treated as a confined space until proven otherwise.	☐ Space classified ☐ Uncertainty = confined space
Entry without a permit; personnel unaccounted for	A Confined Space Entry Permit must be completed prior to any entry. The permit must identify the location and description of the space; list all authorised personnel entering; define the duration of entry; document all identified hazards and control measures; include atmospheric test results; and record sign-in and sign-out of all personnel. No person is to enter a confined space without a valid, current entry permit. A tracking system — permit log, tag-in and tag-out, or standby person record — must be in place. No person is to remain unaccounted for at any time.	☐ Permit valid & current ☐ Hazards documented ☐ Sign in / out log ☐ Tag-in / tag-out
Lack of training or competency	All personnel involved in confined space work must hold current confined space entry training (RIIWH202E <i>Enter and work in confined spaces</i> or equivalent) and be trained in hazard identification, use of gas monitoring equipment, and emergency procedures. Training records must be maintained and verified prior to site attendance. You must not undertake any confined space work unless you have successfully completed this training.	☐ RIIWH202E verified ☐ Gas monitor competency ☐ Records held
Inadequate planning leading to incident or injury on site	Conduct a risk assessment to identify potential hazards. Check the condition of all equipment including harnesses, atmospheric monitoring equipment, hard hat, two-way radios, fire extinguisher, respirators and gloves. Check the first aid kit. Ensure all personnel are trained, accredited, supervised and aware of the risks. Obtain all necessary permits and approvals. Review the site emergency response plan. Review the field plan and perform a communication check-in with all relevant onsite stakeholders and the EC office.	☐ Risk assessment done ☐ Equipment checked ☐ First aid kit checked ☐ Permits obtained ☐ Comms check-in
Atmospheric testing, monitoring and ventilation		
Oxygen deficiency or enrichment; toxic gases; flammable atmosphere	Atmospheric testing must be undertaken by a competent person using calibrated gas detection equipment. Pre-entry testing must be completed from outside the confined space where possible and must test for oxygen (19.5%–23.5%), flammable gases (below 5% LEL) and toxic gases including carbon monoxide and hydrogen sulphide. During entry, continuous atmospheric monitoring must be maintained. Work must cease immediately if oxygen falls below 19.5% or exceeds 23.5%, gas levels exceed safe limits, or alarm thresholds are triggered.	☐ Pre-entry test from outside ☐ O ₂ 19.5–23.5% ☐ LEL below 5% ☐ Toxic gases tested ☐ Continuous monitoring
Lack of ventilation	Ensure adequate ventilation to maintain safe oxygen levels. Use mechanical ventilation where natural ventilation is insufficient, and confirm it is operating before and throughout entry. Re-test the atmosphere after any interruption to ventilation.	☐ Ventilation confirmed ☐ Mechanical ventilation ☐ Re-test after interruption
Ignition of a flammable atmosphere; exposure from prolonged occupancy	Avoid using ignition sources such as open flames or electrical equipment that can spark; use intrinsically safe torches and equipment. Implement safe handling and storage procedures for flammable materials. Limit the time spent in the confined space to reduce exposure.	☐ No ignition sources ☐ Intrinsically safe lighting ☐ Exposure time limited
Standby person, roles and communication		

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Loss of contact; entrapment; delayed rescue; lone-worker risk	A dedicated standby person must be positioned outside the confined space at all times during entry. The standby person must maintain continuous communication with workers inside the space, monitor environmental and worker conditions, have immediate access to rescue and first aid equipment, perform no other task, and must not enter the confined space under any circumstances . Continuous communication must be maintained by voice, radio or hand signals, with regular check-ins and an established emergency escalation procedure.	☐ Standby person posted ☐ No other duties ☐ Continuous comms ☐ Regular check-ins
Unclear roles leading to miscommunication and injury	When discussing the field plan the Field Lead assigns onsite roles and responsibilities to each individual — Site Supervisor (ensures all safety measures are implemented and followed), Workers (follow the SWMS and use PPE as required), Safety Officer (reviews and monitors site WHS compliance), and Communication Officer (conducts regular check-ins with all staff and reports to the EC office every one to two hours).	☐ Roles assigned ☐ Responsibilities briefed ☐ 1–2 hourly office reports
Isolation, biological, water and environmental hazards		
Uncontrolled energy, plant, water inflow or traffic above the space	All potential hazards must be isolated prior to entry, including water inflows or discharge sources, mechanical or electrical systems, adjacent plant or equipment, and vehicle and traffic hazards where working under bridges or in road reserves. Confirm lock-out and tag-out is applied and verified. Where isolation is not possible, risk must be minimised through alternative controls documented on the permit.	☐ Water isolated ☐ LOTO verified ☐ Plant isolated ☐ Traffic controls in place
Biological exposure — bacteria, fungi, contaminated water and animal waste including guano	Use gloves, coveralls and appropriate respiratory protection. Treat all standing water, sediment and surfaces as contaminated. No eating, drinking or smoking in the space. Apply hand hygiene and decontamination procedures after work — remove PPE in a controlled manner, do not shake or blow down clothing, double-bag disposable PPE and contaminated waste, and clean boots and equipment before leaving.	☐ PPE worn ☐ No eat / drink ☐ Hand hygiene ☐ Waste double-bagged ☐ Equipment decontaminated
Fauna hazards — bats, snakes, rodents and insects within the space	Do not disturb fauna unnecessarily. Maintain awareness of animal behaviour and withdraw if an animal is agitated or blocking egress. Only trained personnel handle wildlife where required, under the applicable licence and in accordance with SOP — Terrestrial Fauna. Do not handle bats unless vaccinated and authorised; any bat bite or scratch requires immediate wound washing and urgent medical attention.	☐ Fauna not disturbed ☐ Trained handler only ☐ No unvaccinated bat contact
Flooding, sudden water release and flash flooding	Before entering culverts, drains or water structures, assess upstream rainfall and weather conditions and identify the potential for sudden water release or flash flooding, including dam and detention basin discharge. Do not enter during or after rainfall events where flooding is possible . Monitor water levels continuously during entry and establish an immediate evacuation trigger.	☐ Upstream assessed ☐ No entry after rain ☐ Levels monitored ☐ Evacuation trigger set
Slippery surfaces, unstable substrates and poor lighting	Be aware of slippery surfaces, unstable substrates and poor lighting. Ensure adequate lighting is provided for all works — carry a minimum of two independent light sources per person. Move slowly, maintain stable footing and avoid sudden movements. Do not proceed beyond line of sight unless authorised on the permit.	☐ Two light sources each ☐ Footing stable ☐ Line of sight maintained
Rescue, site clean-up and close-out		

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Entrapment or injury with no rescue capability; delayed rescue	A site-specific rescue plan must be developed prior to entry and must be based on non-entry rescue methods wherever possible, including harness and lifeline and a tripod or retrieval system where applicable, and must ensure rescue can be initiated immediately without delay. Emergency contacts and nearest medical facilities must be identified, access routes for emergency services confirmed, and all personnel briefed on emergency procedures prior to entry. Ecology Consulting staff must not attempt confined space rescue — rescue is conducted by the Principal Contractor or emergency services.	☐ Rescue plan (non-entry) ☐ Retrieval system rigged ☐ Medical facilities identified ☐ All briefed pre-entry
Contaminated material and waste left on site	Properly dispose of any contaminated materials in accordance with site waste procedures. Ensure the site is clean and free of hazards before leaving, and confirm all personnel and equipment are accounted for and the space is secured.	☐ Waste disposed ☐ Site left clear ☐ Headcount complete

Stop-work triggers — stop work immediately, move to a safe location, notify the Field Lead and Principal Contractor, and do not recommence until the risk is controlled and communicated			
Oxygen outside 19.5–23.5% or gas alarm triggered	Ventilation fails or is interrupted	Rising water levels or rainfall upstream	Loss of communication with the standby person
Structural instability observed	Lighting inadequate or failed	Permit expired or conditions changed	Any unidentified hazard present

Emergency response

Read with the site-specific Emergency Response Plan. The Field Lead is the Chief Warden. Any worker may activate the ERP.

1 · Activate & make safe	Stop work and ensure the area is safe for responders. Do not move an injured person unless there is immediate danger. For a confined space incident, initiate non-entry rescue immediately using the retrieval system. Ecology Consulting personnel must not enter the space to attempt a rescue — the Communication Officer initiates the emergency response and rescue is performed by the Principal Contractor or emergency services. The Field Lead assesses the situation, initiates the response, directs evacuation to the Emergency Meeting Point, accounts for all personnel using the Daily Toolbox Talk sign-on sheet, and coordinates communication with the office.
2 · Call for help	Call 000 or 112, or activate the In Reach or PLB where there is no coverage. Provide GPS coordinates and the nearest cross roads, the nature of the emergency and safe access routes. Assign one person to guide responders in.
3 · First aid	A trained first aider assesses the casualty, but only where it is safe to do so. Control bleeding, immobilise suspected fractures and monitor breathing. If unconscious but breathing, place in the recovery position; if not breathing, commence CPR. Record all treatment on the First Aid Treatment Log.
4 · Post-incident	Report to the WHS Officer and complete an Incident Report Form within the required timeframe, submitted through the digital WHS platform. The WHS Officer updates the Incident Register and notifies the regulator of any notifiable incident. Attend debrief; review and update this SWMS and the ERP; restock emergency equipment used. Psychosocial support is provided where workers were exposed to distressing events.

Emergency services	000 or 112	State Emergency Service (SES)	13 25 00
EC office / Check-in Officer	0472 569 988	Managing Director / WHS Officer	Tony Webster · 0448 439 334
Poisons Information Hotline	13 11 26	Wildlife rescue	1300 094 737
Employee Assistance Program	1300 172 327 — free and confidential	Nearest hospital & muster point	Identified per site in the ERP; confirmed at the toolbox talk

Training & competency requirements

Required for this activity	RIIWHS202E <i>Enter and work in confined spaces</i> or equivalent is mandatory for all entrants and the standby person, together with training in hazard identification, gas monitoring equipment and emergency procedures. Gas detectors must be within calibration date and bump-tested before use. Working at Heights training is required where a harness and retrieval system is used.
Baseline — all field personnel	White Card (General Construction Induction); current driver's licence of appropriate class; CPR certification (annual renewal); Provide First Aid (HLTAID011), and Remote First Aid (HLTAID013) for remote or isolated work; General Fauna Handling Training.
Verification	All training must be current, recorded and verifiable prior to mobilisation. Records are maintained by the Administration Officer and retained for a minimum of five years, with expiry notifications issued in advance; workers are not mobilised with expired mandatory training. Competency is verified annually and observed on site by Principal or Senior Ecologists, Project Managers and the WHS Officer. Subcontractor training evidence is collected and retained for audit.

Daily SWMS acknowledgement & sign-on register

All personnel must read, understand and sign this SWMS before commencing work, confirming they are fit for duty, competent and able to perform assigned tasks safely. Completed with the Daily Toolbox Talk and submitted through the digital WHS platform.

#	Name	Role	Signature	Date
1				
2				
3				
4				
5				
6				

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